UFSO

SCIENCE OLYMPIAD
AT THE
UNIVERSITY OF FLORIDA

Northern Regional: January 18th, 2020

Water Quality C

Name(s): _____

Team Name: _____

School Name: _____

Team Number: _____

Rank: _____

Score: _____

Part I. Marine and Estuary Ecology

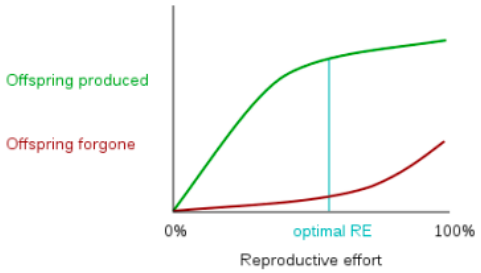
(30 Points)

1. What is the term for an excess of nutrients in a freshwater ecosystem? (1)

2. List four things that stormwater runoff transports that can be harmful in freshwater ecosystems. (4)

3. Freshwater biomes supply how much of the world's freshwater? (1)

   A. 10%
   B. 33%
   C. 50%
   D. 25%
   E. 90%



4. Refer to the above graph.

   a. What type of reproduction is modeled by the graph? (1)

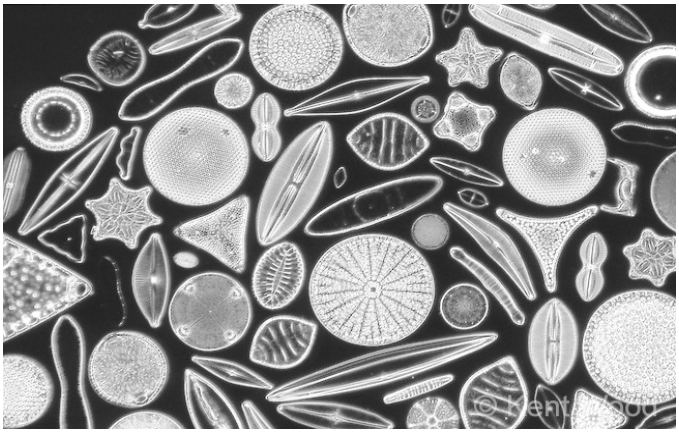
   b. What determines the "optimal RE"? (1)

   c. (True/False) This type of reproduction is consistent with that of a salmon. (1)

   d. Is this type of reproduction consistent with R or K selection? (1)

5. Define an invasive species (3) and give an example of a marine invasive species (common name is fine) (1).

Questions 6-8 refer to the below picture of diatoms, a major group of algae.



6. What is a diatom's skeleton called? (1)

a) shell

b) frustule

c) cover

d) exoscutum

7. How would you most likely see diatoms with the naked eye? (1)

a) coating a rock or hard surface

b) swimming in schools

c) colonies on the seabed

d) they are microscopic and cannot be seen with the naked eye

8. What is the main feature of diatom's shape? (1)

a) branching

b) symmetry

c) diamond

d) tubular

9. Which of these are NOT a habitat commonly found in estuaries? (1)

a) mangrove swamp

b) oyster reef

c) kelp forest

d) seagrass bed

e) coral reef

10. Which of the following invertebrates is NOT primarily benthic? (1)

a) sponges

b) mussels

c) bryozoans

d) copepods

e) polychaetes

11. Which of the following is NOT a key abiotic factor in marine ecosystems? (1)

a) water temperature

b) water depth

c) sunlight availability

d) plankton availability

12. The place where the ocean meets the land is the (1)

a) intertidal zone

b) neritic zone

c) benthic zone

d) oceanic zone

13. Identify the term used to describe a population whose growth is now limited by various limiting factors: (1)

a) exponential decline

b) density-dependent growth

c) logistic growth

d) immigration growth

14.What percent of all plants and animals in the ocean live around coral reefs? (1)

a) 25

b) 30

c) 10

d) 50

15. Define the following processes of the water cycle. (3)

a. Precipitation intercepted by foliage that is evaporated

b. Movement of water through the atmosphere

c. Flow of water in the vadose zone and aquifers

16. Overfishing can artificially introduce _____ into a food web (1)

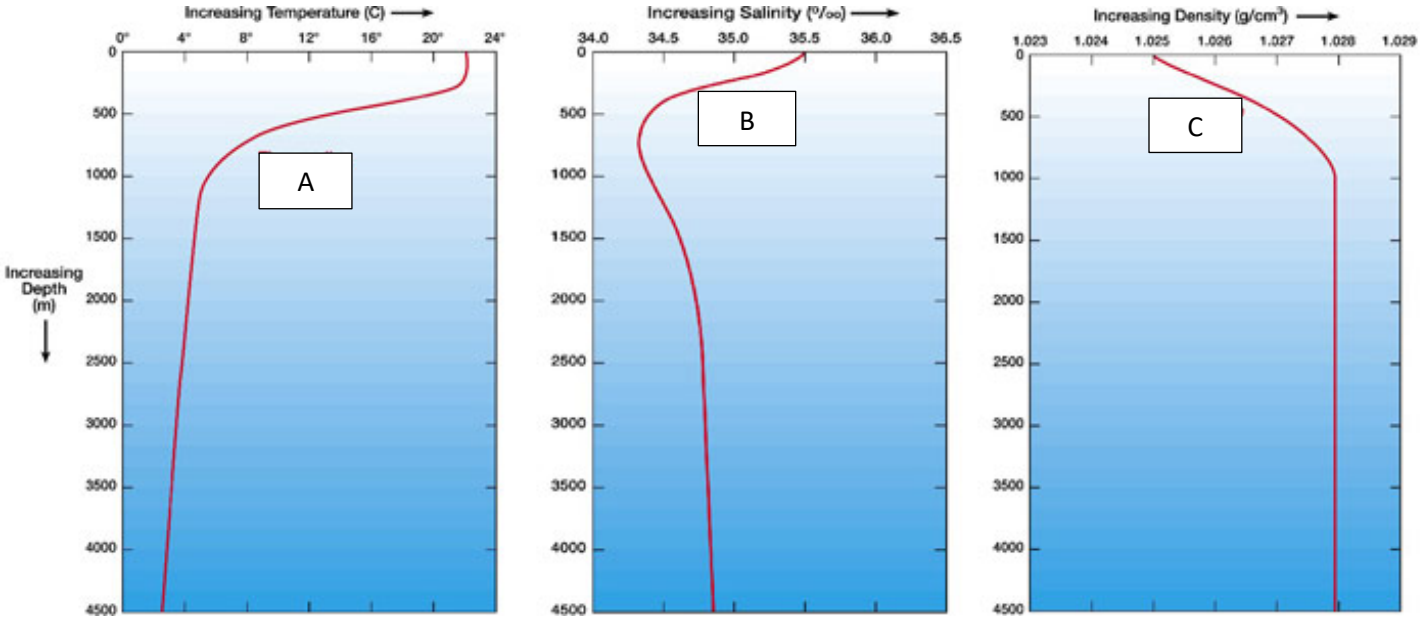
a) Top-down control

b) Pollution

c) Trophic cascades

d) Nutrient spiraling

Questions 17-19 refer to below photo. Match the corresponding letter on the graph to the term if refers to.



17. Halocline (1)

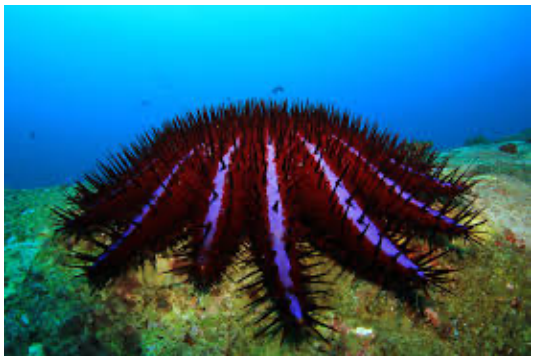
18. Pycnocline (1)

19. Thermocline (1)

Part II. Coral Reef Macroflora and Fauna Identification

(30 Points)

A



B



C



D



E



F



Questions 1-6. Refer to the above pictures. Each letter corresponds with the organism pictured below it.

1. Which of these organisms is found only in the Atlantic region? (1)

2. Which of these organisms would be found in the Indo-Pacific region but never the Atlantic? (3)

3. The following questions refer to organism A:

a. What is the common name? (1)

b. This organism primarily consumes ____? (1)

c. What is this species conservation status? (1)

4. Identify which fish has the longest lifespan. (1)

5. Identify which of these organisms has the longest lifespan (could be same answer as above). (1)

6. Indicate whether or not each organism exhibits hermaphroditism (Yes/No). (6)

A

B

C

D

E

F

7. Corals are members of phylum _____. (1)

8. Briefly explain how coral bleaching occurs and how it damages corals. (3)

9. Briefly explain how ocean acidification occurs and how it damages corals. (3)

10. Hermatypic corals differ from ahermatypic corals in that hermatypic corals _____. (1)

11. Put the following in order from youngest to oldest: atoll, fringing reef, barrier reef. (1)

Specimen	Species	Biomass (g)
1	A	250
2	B	35
3	B	42
4	B	41
5	B	37
6	C	79
7	C	82
8	C	87
9	D	120
10	E	170
11	F	100

Questions 12-14. Refer to the above chart. 11 fishes were caught at the SciOly Coral Reef Research Center.

12. What is the species richness? (1)

13. What is the abundance? (1)

14. Which species has the greatest overall biomass? (1)

15. Name any 3 of the 6 countries that make up the core of the Coral Triangle. (3)

Part III. Water Monitoring Analysis

(30 Points)

Questions 1-4 test your knowledge on different chemical analysis tests for water quality.

1. Temperature affects the amount of gases such as oxygen that can be dissolved in the water. Cold water holds (more/less) oxygen than warm water. Temperature

(increases/decreases) metabolic rate of aquatic organisms. Raising water temperature (increases/decreases) decomposition rate of organic matter. (3)

2. pH is measured with a pH (tool) _____. Normal pH of rivers in the US is between 6.5 and ____. Acid rain has a pH of less than ____. (3)

3. _____ measures the clarity or haziness of water in a given body. (1)

4. Identify the most appropriate tool to measure and analyze each of the following indicators.

a. Pesticides and organic pollutants (1)

b. Heavy metals (1)

c. Total dissolved solids (1)

5. Which of these common dissolved solids are not essential nutrients for life. (1)

A) magnesium

B) nitrogen

C) phosphorus

D) sulfur

6. Identify the aquatic life that can survive at the lowest pH levels. (1)

A) snails, clams

B) carp, catfish

C) algal or rooted plants

D) crustaceans

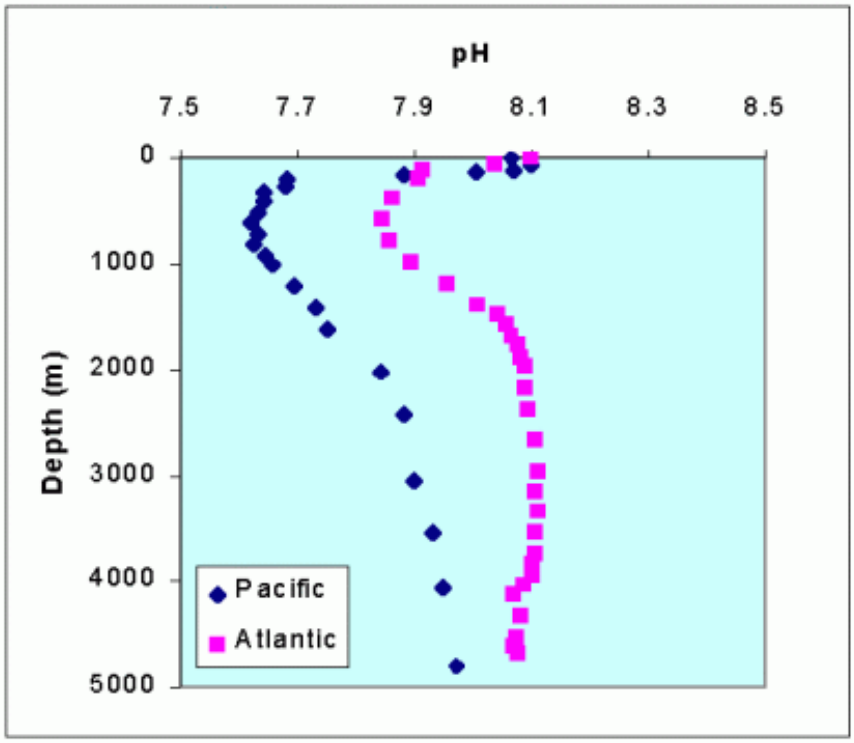
E) mayfly, dragonfly nymph

7. Identify natural factors that may affect dissolved oxygen levels. (8)

8. Identify human factors that may affect dissolved oxygen levels. (4)

9. Suppose from a sample of 250 mls you obtain 120 mg of dried residue. How many mg of total solids per liter are in the sample? Show your work. (2)

10. Freshwater systems usually have _____ levels of phosphorus compared to estuaries. (1)
    a. Higher
    b. Lower
    c. The same



Questions 11-13 Refer to the above graph.

11. Between what depth does pH begin increasing? (1)

12. In which ocean does pH initially decrease more rapidly with depth? (1)

13. Which ocean would you expected to have more dissolved gases in the deepwater? (1)

Part IV. Salinometer/Hydrometer

(10 Points)

1. This final section requires you to demonstrate the use of your homemade salinometer. (10)
    a. Show your salinometer to the event supervisor or designated volunteer and have them initial your answer sheet. (5)
    b. You may wish to calibrate your salinometer prior to testing the unknown solution.
    c. Record the concentration of solution A to the nearest tenth of a percent, for example, 8.5%. (5)

Tie Breakers

Only graded in event of a tie.

1. Most estuaries are not called estuaries, and often have other, sometimes misleading, names. List as many other terms for estuary that you can think of.

2. Identify the common or scientific name of the below organism.



3. List as many coral diseases as you can think of.